

NETCONF and YANG

1 The Network Configuration Protocol (NETCONF)

NETCONF provides messaging to retrieve, set, and modify configuration data at managed devices. It moves away from the traditional header-based protocol formats (like SNMP) to a structured **XML document** approach.

Core Paradigm

- **paradigm:** Uses Remote Procedure Call (RPC) for communication.
- **Encoding:** Messages are encoded in **XML**.
- **Transport:** Operates over secure, connection-oriented sessions like **TLS over TCP**.

2 NETCONF Sessions and Operations

A NETCONF session involves several steps: establishing a secure connection, exchanging **capabilities** (via <hello> messages), and then performing RPC operations.

Operation	Description
<get-config>	Retrieve all or part of a specified configuration (e.g., the running config).
<get>	Retrieve configuration and operational state data.
<edit-config>	Change a device's configuration. Supports rollback on error.
<lock> / <unlock>	Short-lived locks to prevent conflicting changes from other managers.
<notification>	Asynchronous event notifications sent from agent to manager.

Table 1: Selected NETCONF Operations.

Atomic Transactions

NETCONF allows for multi-device transactions that either complete successfully as a group or are fully reversed. This enables management of the **network as a whole** rather than individual devices.

3 XML Examples

Retrieving All Data (<get>)

```
<rpc message-id="101" xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
  <get/>
</rpc>
```

The device responds with a <rpc-reply> containing the full configuration data in XML.

Editing configuration (Set MTU)

Setting the MTU of interface Ethernet0/0 to 1500:

```
<rpc message-id="101" xmlns="urn:ietf:params:xml:ns:netconf:base:1.0">
  <edit-config>
    <target><running/></target>
    <config>
      <interface>
        <name>Ethernet0/0</name>
        <mtu>1500</mtu>
      </interface>
    </config>
  </edit-config>
</rpc>
```

4 YANG

YANG is the data modeling language used to specify the structure, syntax, and semantics of network management data for NETCONF.

Key Features of YANG

- **Constraints:** Expresses powerful rules that must be satisfied for a configuration to be valid.
- **Modules:** Definitions are organized into modules for reusability and clarity.
- **Consistency:** Helps ensure that configurations across DIFFERENT devices follow the same structure and rules.

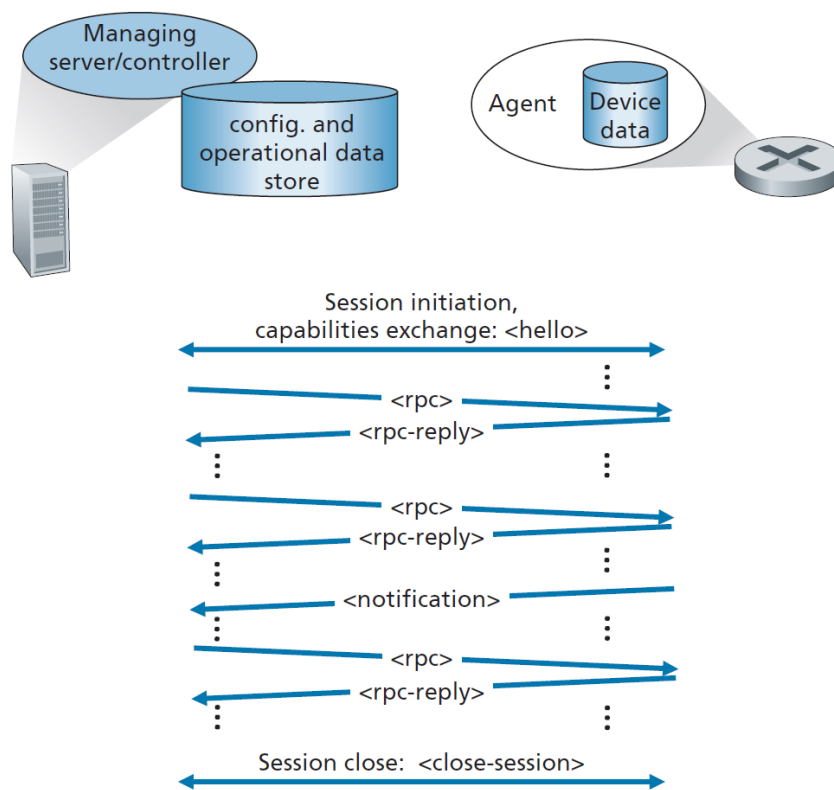


Figure 1: NETCONF Session: Capabilities exchange, RPCs, and notifications.